Self-driving cars project



HOW TECHNOLOGIE CAN HELP

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Module: Computational Thinking

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Introduction

- Self-driving cars will make transportation easier. Car companies and software companies are working together to improve self-driving technology. At the same time, there are many software projects that help to find solutions for self-driving cars which make it easier for others to develop their applications (Cha and Lee, 2021).
  In fact, self-driving cars relies all on technology by using python programming, sensors or even the communication that happens between two cars.
  In my project I created a robot that represent the self-driving cars in a theoretical world using python programming, and showing how this project could save the world problem which is the accidents.
  Indeed, it's important to explore how self-driving can reduce traffic accidents by identifying the type of collisions and human mistakes. For example (Petrović, Mijailović and Pešić, 2020), made a comparison between the autonomous vehicle and the regular ones, and they found that the most typical collision for autonomous vehicle is the "rear-end", since the autonomous cars stops suddenly when it detects the danger and if the regular car or the human who's driving didn't respect the safety distance it's going to crash. And this happens more often than the other collisions including "pedestrian" and "broadside" due to human mistake of not respecting the speed or the distance.
  However, there is a lot to explore about the self-driving car including how to communicate, sense, react, avoid, reroute, resolute and how it applies the smooth operation, by improving these systems self-driving cars will be able to reduce traffic jams and make road safer.

The Problem

- Despite the advantages of self-driving cars there are a lot of problems related to it. (Boosarapu, Mounika, and Shobha Rani, 2021), self-driving cars contain a big amount of data, they need to sense, see and understand. Which makes taking decisions more complex especially in serious and dangerous situations, when it should choose who to prioritize, and it makes it more difficult to create a safe and effective self-driving car in all situations. We should take the risk of self-driving cars more seriously as sometimes the car itself could lose control when it fails to detect an obstacle, with that the driver should stay aware of the car behavior in the end it's just a machine programmed and made to help humans to avoid mistakes, but it can never replace the human brain.
  However, to clarify the purpose of this project we should understand why self-driving cars should replace regular ones, since the regular cars are guided by human brains only which is not enough to reduce accidents as human errors are the principal reason for it. But, with self-driving cars we can unite human brains with technology as it's going to help in a wide way to minimize the risk using camera, sensors and communication. In addition, it provides mobility for people who cannot drive like elderly people or disabled.
  In fact, self-driving cars are becoming a reality. (Davidson and Spinoulas, 2015), we can see this progress with many companies like tesla and google but still facing some challenges like laws, safety and public acceptance which is needed to be solved since the impact of these vehicles will play a big role in shaping the future transportation.

THE ROLE OF SENSORS

- Sensors in self-driving cars help the vehicle detect and understand. (Woodward and Kilestik, 2021, p. 52), highlight that autonomous vehicle perception sensor data and decision-making algorithms can make it possible for a better navigation and a better traffic management, which in turn can decrease traffic accidents significantly.

  Each sensor plays a critical role for example we have the “LIDAR” sensor that use the laser and calculate how long it takes for the light to return after hitting an object to detect the distance to create a 3D map (Li et al., 2022). The “Ultrasonic sensors” this one detects nearby objects at low speeds using the sound ways with the same way that LIDAR sensor does, and it’s usually used during parking (Kelemen et al., 2015). The “cameras” is like how humans use their eyes, can detect movement, shapes and colors it provides cars to identify road signs and the traffic lights (Nakamura, 2017). The “Radar” this one use radio waves to detect objects and measure their speed and distance especially in bad weather (Yan et al., 2022). And there is a lot to mention.

  These sensors aligns perfectly with my project in webots I used sensors to detect the object and avoid it, specifically I used IR sensors used for proximity detection which is nearly similar to the “Ultrasonic sensor” as it detects nearby obstacles but the key difference is this one uses sound and the IR sensor uses light, however at this point it aligns with the LIDAR sensor as both uses lights but in different way which makes LIDAR sensor more advanced.

  Nevertheless, my robot was programmed with python to avoid obstacle and decide whether to stop or turn, and the purpose was to show how this theoretical project could solve the car accidents by showing how cars can interact with each other, communicate, sense and react.

PROGRAMMING AND ALGORITHMS

- Programming and algorithms are necessary to create a robot represent self-driving cars behaviors.

  To create the robot, I used webots app. At first, I designed my robot with the shapes and the motors then I created the sensors and the GPS, I put everything in the scene tree but indeed this is not enough to say that we created a robot. The most important part is the controller, I created a controller to control the robot behavior by writing algorithms.

  The first algorithm I started with was the motors initialization (appendix A_1). Then, the sensors initialization and enable with the timestep to start collecting data at each simulation step (appendix A_2). After that I created a collision detection to detect the collision by checking the left and right sensors to detect an obstacle if their values are bellow 800, it means if both sensors detect low value means there is an obstacle in the front the robot then adjust the motors speed and stop (appendix A_3). And to make it clear I made an obstacle avoidance “when obstacle detected on the right turn left, when obstacle detected on the left turn right” it means when an obstacle detected on one side the motor slows on that side and speed on the opposite one to turn to the proper direction, and to make this move I made the robot to calculate if the right or left sensor are below the thresholder value which is the value used as boundary to compare with the IR sensors(appendix_4). To make the robot move I made an algorithm to move forward if no obstacle detected , the wheels in this case are set to the same speed(appendix A_5).However, the motors won’t move if we didn’t adjust the speed and that’s why I made an algorithm for that to calculate the motors speed based on the computed values from the loop of obstacle avoidance (appendix A_6). And finally, I made a GPS monitor algorithm to calculate the position of the robot and print them (appendix A_7).

  Some of these algorithms are made in a loop to track obstacles and avoid them (appendix A_8).

  The robot was designed with a square shape and four wheels, adding four sensors in the front to detect obstacles (appendix A_9). Finally, when I ran my robot it seems to act perfectly.

CAR COMMUNICATION

- Not widely known but in fact self-driving cars uses communication with many different entities, it can be a car with a car or a car with objects.

  The communication could be with the roads, weather updates or even the traffic lights where it happens an exchange of signals to sign the cars it's going to be red or green, the most common one is the car with pedestrian many cars nowadays contain a sensor in the back for parking and a camera to detect people cars or anything else for advanced ones according to the U.S. Department of Transportation (2022), the sensors could detect people by their smartphones the car could adjust the speed by detecting the smartphone signal to avoid accidents, in addition the car could communicate with another car In the road to avoid crashes by sharing data.

  We can say that it's the same things happening in my project where the robot communicates with the object or the walls through the sensors and while using the GPS it communicates with the floor to look for the proper coordination given in the controller code which is the destination.

  And just like the self-driving cars in real world makes decisions, also my robot makes decision when it detects the obstacle, he decides whether to turn right, left or stop. Also, my robot has been tested many times to control the movement and to adjust the problems in the simulation world.

COMPLEX PROBLEM-SOLVING

- I can’t deny while I was preparing my project I face I lot of challenges with my little knowledge of how to use webots, there are many problems to mention starting with creating the robot I was struggling at first to make a shape or a wheel but after I understood how it works I made my own car with my own way and indeed I used some help by visiting the `Webots User Guide provided by Cyberbotics Ltd. (2023).`

  After I finished the design of the robot the second thing to think about is how to make a controller using the help of the same cite again while it shows what algorithms can you use and how to use them, moving to the sensors part where I searched a lot to find how to make a sensor while it was super easy. After the sensors it comes the algorithm that we going to put into the controller to control the robot moves, after trying a lot of codes I finally found the proper logical way to use for the robot is the if loop where it acts on what the sensors detected. However, sometimes when I run the simulation the robot won’t move then I check the motors name if I called it with the proper name or not and sometimes I found that the robot goes to the opposite way while when I look for the problem where is it, I find that the max_speed should be -max_speed for my robot, and the hardest part for me was to find how to let the robot reach a specified destination.

Conclusion

- Self-driving cars represent a massive technology advancement gathering artificial intelligence, sensors and machine learning to make it safer to solve the transportation problems, however we mentioned the self-driving cars can make mistakes sometimes and for that we need the human brain to help in this case with this issue to make decisions.

  In addition, we can't deny that the sensors play a big role in self-driving cars technology they are almost everything, the car it relies on how the sensor detect the obstacle and act to avoid it all of this should be written in a controller section with algorithms to activate the sensors and the motors. In fact, the car makes a communication with the environment to understand, make decisions and act in the proper time by detecting the signs.

  my project represents a simulated world where it shows the self-driving cars how it could react avoid, make decisions and communicate

  Finally, it's true that I faced a lot of problems to create that project but also, I learned a lot of things through it, I learned how to work with a new application, how to write algorithms in proper way, and made me curious to learn more about self-driving cars.

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Appendices

- Appendix A_1:

```
left_front_motor = robot.getDevice('motor_1')
left_rear_motor = robot.getDevice('motor_2')
right_front_motor = robot.getDevice('motor_3')
right_rear_motor = robot.getDevice('motor_4')

left_front_motor.setPosition(float('inf'))
left_rear_motor.setPosition(float('inf'))
right_front_motor.setPosition(float('inf'))
right_rear_motor.setPosition(float('inf'))

left_front_motor.setVelocity(0.0)
left_rear_motor.setVelocity(0.0)
right_front_motor.setVelocity(0.0)
right_rear_motor.setVelocity(0.0)
```

- Appendix A_2:

```
 # Initialize the IR sensors
ir_left = robot.getDevice("ir_left")
ir_left_1 = robot.getDevice("ir_left_1")
ir_right = robot.getDevice("ir_right")
ir_right_1 = robot.getDevice("ir_right_1")
ir_front = robot.getDevice('ir_front')

#enable the IR sensors with timestep
ir_left.enable(timestep)
ir_left_1.enable(timestep)
ir_right.enable(timestep)
ir_right_1.enable(timestep)
ir_front.enable(timestep)
```

- Appendix A_3:

```
if ir_left_value < 800 and ir_right_value < 800:
    print("Collision detected! Avoiding or Stopping.")
    left_speed = 0.0
    right_speed = 0.0
```

- Appendix A_4:

```
elif ir_left_value < ir_thresholder:
    print("Object detected on the left. Turning right.")
    left_speed = max_speed * 0.2
    right_speed = max_speed * 0.5
elif ir_right_value < ir_thresholder:
    print("Object detected on the right. Turning left.")
    left_speed = max_speed * 0.5
    right_speed = max_speed * 0.2
```

- Appendix A_5:

```
else:
    print("No object detected. Moving forward.")
    left_speed = 0.5 * -max_speed
    right_speed = 0.5 * -max_speed
```

- Appendix A_6:

```
left_front_motor.setVelocity(left_speed)
left_rear_motor.setVelocity(left_speed)
right_front_motor.setVelocity(right_speed)
right_rear_motor.setVelocity(right_speed)
```

- Appendix A_7:

```
gps_value = gps.getValues()
#print(gps_value)

msg = "GPS values: "
for each_val in gps_value:
    msg += " {0:0.5f}".format(each_val)
print(msg)
```

- Appendix A_8:

```
if ir_left_value < 800 and ir_right_value < 800:
    print("Collision detected! Avoiding or Stopping.")
    left_speed = 0.5
    right_speed = 0.0
elif ir_left_value < ir_thresholder:
    print("Object detected on the left. Turning right.")
    left_speed = max_speed * 0.2
    right_speed = max_speed * 0.5
elif ir_right_value < ir_thresholder:
    print("Object detected on the right. Turning left.")
    left_speed = max_speed * 0.5
    right_speed = max_speed * 0.2
else:
    print("No object detected. Moving forward.")
    left_speed = 0.5 * -max_speed
    right_speed = 0.5 * -max_speed
```

- Appendix A_9:

